16	**(i)**		$C = 2$ $A = 62$ $B = 10$	**B1** **B1** **B1** **[3]**	**3.3** **3.3** **1.1**	since max when $t = 2$ since max when $(t-2)^2 = 0$ from substitution of 22, 62 and 2	
16	**(ii)**		substitution of 0.75 in $p = 62 - 10(t-2)^2$ 46	**M1** **A1** **[2]**	**3.4** **1.1**	FT *their* 2, 62, 10 allow 46.375 rounded to 2 or more sf	

Question			Answer	Marks	AOs	Guidance	
16	(iii)		*their* $62 - 10(t-2)^2 = 0$	**M1**	**3.4**	or ≥ 0 or > 0 for **M1** NB $t = 2 + \sqrt{6.2}$	
			$[t =]$ 4 hours 29 minutes or 4 hours 30 minutes	**A1** **[2]**	**2.4**	allow 4.49 or 4.5 [hours]	
16	(iv)		substitution of $t = 1$, 3 and 5 awrt $59.4 \approx 59$ awrt $83.8 \approx 84$ awrt $88.8 \approx 89$	**M1** **A1** **[2]**	**3.4** **3.5a**	or awrt 59.4, 83.8 and 88.8 found and supporting comment made eg they are approximately the same as the values in the table	if **M0** allow **SC1** for two values correctly found and shown to be consistent or supporting comment made
16	(v)		$p \rightarrow 90$ as $t \rightarrow$ large or when $t = 12$ $p = 89.99539\ldots$rounded to 2 or more sf comparison with value of p for $t = 5$ eg model predicts $p = 89$ for $t = 5$ and $p = 90$ for $t = 12$ so not good advice	**B1** **B1** **[2]**	**3.5a** **3.5a**	or model predicts $p = 90$ for (any) $t \geq 7$ so not good advice	allow equivalent comment on 7 hours work for one extra mark